

Databases

Laboratory Work 1

ER-Diagram of the International Airport Database

Student: Merei Mukhametkali

Tool used: PostgreSQL / IntelliJ IDEA

Submission: ER-Diagram and textual solution description

1. System Description Summary

The International Airport Database manages core airport operations: airport and airline information, scheduled and actual flights, passenger profiles, bookings, boarding passes, baggage registration and checks, security checks, and a separate history of booking flight changes. The schema consists of ten related tables connected by primary-key and foreign-key constraints.

Database structure: 10 entities, primary keys for entity identification, foreign keys for relationships, UNIQUE constraints for airline and passport identifiers, and timestamp fields for creation and update tracking.

Modeling approach: each operational concept is represented by its own table. Relationship cardinality is determined directly from the foreign-key nullability and UNIQUE constraints present in the current PostgreSQL schema.

Scope of the Laboratory Work

The report focuses only on the ten tables implemented in the current database. No additional entities, attributes, or business rules are added beyond the schema.

2. Entities and Attributes

airport

Stores airports and their geographic location and audit timestamps.

Attribute	Data type	Key / constraint	Description
airport_id	SERIAL	PK	Unique identifier of the airport
airport_name	VARCHAR(100)	NOT NULL	Full name of the airport
country	VARCHAR(50)	-	Country where the airport is located
state	VARCHAR(50)	-	State or region of the airport
city	VARCHAR(50)	-	City where the airport is located
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Record last-update timestamp

airline

Stores airline companies and their identifying code.

Attribute	Data type	Key / constraint	Description
airline_id	SERIAL	PK	Unique identifier of the airline
airline_code	VARCHAR(10)	UNIQUE, NOT NULL	Unique airline code
name	VARCHAR(100)	NOT NULL	Airline company name
country	VARCHAR(50)	-	Country of the airline
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Record last-update timestamp

2. Entities and Attributes (continued)

passenger

Stores passenger identity, citizenship, residence, passport, and audit data.

Attribute	Data type	Key / constraint	Description
passenger_id	SERIAL	PK	Unique identifier of the passenger
first_name	VARCHAR(50)	NOT NULL	Passenger's first name
last_name	VARCHAR(50)	NOT NULL	Passenger's last name
gender	VARCHAR(10)	-	Passenger gender
date_of_birth	DATE	-	Passenger date of birth
country_of_citizenship	VARCHAR(50)	-	Passenger citizenship country
country_of_residence	VARCHAR(50)	-	Passenger country of residence
passport_number	VARCHAR(20)	UNIQUE, NOT NULL	Unique passport number
created_at	TIMESTAMP	DEFAULT NOW()	Profile creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Profile last-update timestamp

flight

Stores flight schedule, actual times, gates, airline, and airport references.

Attribute	Data type	Key / constraint	Description
flight_id	SERIAL	PK	Unique identifier of the flight
airline_id	INT	FK, NOT NULL	References airline.airline_id
departure_airport_id	INT	FK, NOT NULL	References airport.airport_id for departure
arrival_airport_id	INT	FK, NOT NULL	References airport.airport_id for arrival
departing_gate	VARCHAR(10)	-	Departure gate
arriving_gate	VARCHAR(10)	-	Arrival gate
scheduled_departure_time	TIMESTAMP	NOT NULL	Planned departure date and time
scheduled_arrival_time	TIMESTAMP	NOT NULL	Planned arrival date and time
actual_departure_time	TIMESTAMP	-	Actual departure date and time
actual_arrival_time	TIMESTAMP	-	Actual arrival date and time
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Record last-update timestamp

2. Entities and Attributes (continued)

booking

Associates a passenger with a flight and stores booking details and price.

Attribute	Data type	Key / constraint	Description
booking_id	SERIAL	PK	Unique identifier of the booking
flight_id	INT	FK, NOT NULL	References flight.flight_id
passenger_id	INT	FK, NOT NULL	References passenger.passenger_id
status	VARCHAR(20)	-	Current booking status
booking_platform	VARCHAR(50)	-	Platform used to make the booking
ticket_price	NUMERIC(10,2)	-	Ticket price
created_at	TIMESTAMP	DEFAULT NOW()	Booking creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Booking last-update timestamp

booking_change

Stores the history of changes associated with a booking, including optional old and new flight references.

Attribute	Data type	Key / constraint	Description
booking_change_id	SERIAL	PK	Unique identifier of the booking change record
booking_id	INT	FK, NOT NULL	References booking.booking_id
change_description	VARCHAR(255)	-	Description of the change
old_flight_id	INT	FK	Previous flight, when applicable
new_flight_id	INT	FK	New flight, when applicable
created_at	TIMESTAMP	DEFAULT NOW()	Change record creation timestamp

2. Entities and Attributes (continued)

boarding_pass

Stores the boarding pass issued for a booking; the booking reference is unique.

Attribute	Data type	Key / constraint	Description
boarding_pass_id	SERIAL	PK	Unique identifier of the boarding pass
booking_id	INT	FK, UNIQUE, NOT NULL	References booking.booking_id; one pass per booking
seat	VARCHAR(10)	-	Assigned seat
boarding_time	TIMESTAMP	-	Boarding time
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Record last-update timestamp

baggage

Stores baggage registered for a booking and its weight.

Attribute	Data type	Key / constraint	Description
baggage_id	SERIAL	PK	Unique identifier of the baggage item
booking_id	INT	FK, NOT NULL	References booking.booking_id
weight_kg	NUMERIC(5,2)	-	Baggage weight in kilograms
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Record last-update timestamp

2. Entities and Attributes (continued)

baggage_check

Stores baggage checking results and links the check to baggage, passenger, and booking.

Attribute	Data type	Key / constraint	Description
baggage_check_id	SERIAL	PK	Unique identifier of the baggage check
baggage_id	INT	FK, UNIQUE, NOT NULL	References baggage.baggage_id; one check per baggage item
passenger_id	INT	FK, NOT NULL	References passenger.passenger_id
booking_id	INT	FK, NOT NULL	References booking.booking_id
check_result	VARCHAR(50)	-	Baggage security check result
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Record last-update timestamp

security_check

Stores passenger security check results and timestamps.

Attribute	Data type	Key / constraint	Description
security_check_id	SERIAL	PK	Unique identifier of the security check
passenger_id	INT	FK, NOT NULL	References passenger.passenger_id
check_result	VARCHAR(50)	-	Passenger security check result
created_at	TIMESTAMP	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Record last-update timestamp

3. ER Diagram

Updated relational ER diagram based on the current PostgreSQL schema. Relationship labels use the notation parent cardinality : child cardinality.

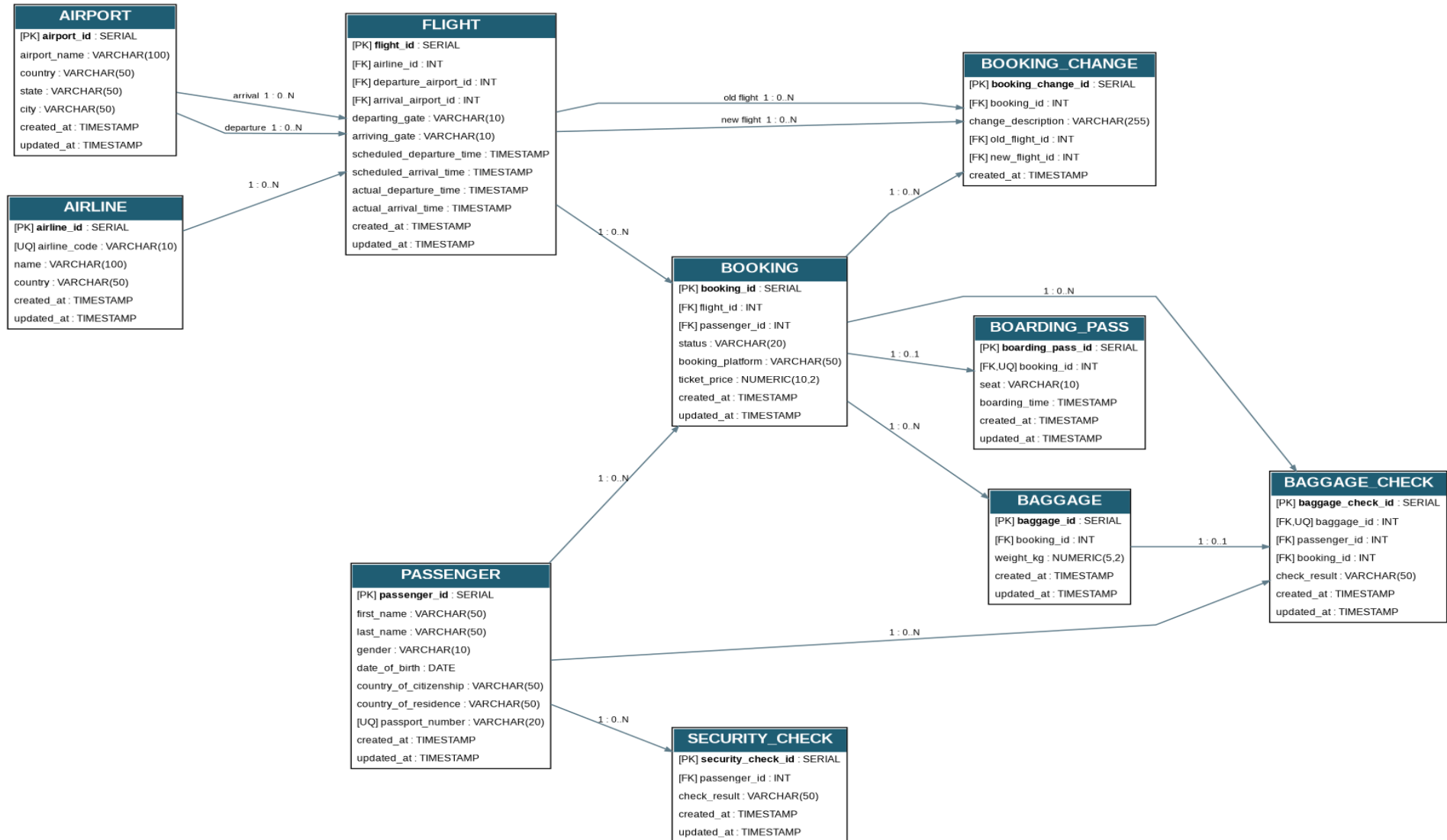


Figure 1. International Airport Database - current schema (10 tables).

4. Legend

Notation	Meaning
PK	Primary Key - uniquely identifies a row in the table.
FK	Foreign Key - references a primary key in another table.
UNIQUE / UQ	Unique value - duplicate values are not allowed.
NOT NULL	Required value - the column cannot contain NULL.
1:M / 1:N	One-to-Many relationship - one parent may be related to many child rows.
1:1 (implemented as 1:0..1)	One parent may have at most one child row when the child FK is UNIQUE; the child must reference exactly one parent.
DEFAULT NOW()	If omitted, PostgreSQL automatically stores the current timestamp.

5. Relationships and Cardinality

Relationship	Cardinality / participation	Implementation (FK)
Airport - Flight (departure)	1 : 0..N Every flight has one departure airport; an airport may be the departure point for many or zero flights.	flight.departure_airport_id → airport.airport_id
Airport - Flight (arrival)	1 : 0..N Every flight has one arrival airport; an airport may be the arrival point for many or zero flights.	flight.arrival_airport_id → airport.airport_id
Airline - Flight	1 : 0..N Every flight belongs to one airline; an airline may operate many or zero flights.	flight.airline_id → airline.airline_id
Flight - Booking	1 : 0..N Every booking is for one flight; a flight may have many or zero bookings.	booking.flight_id → flight.flight_id
Passenger - Booking	1 : 0..N Every booking belongs to one passenger; a passenger may have many or zero bookings.	booking.passenger_id → passenger.passenger_id
Booking - Booking Change	1 : 0..N Every change record belongs to one booking; a booking may have many or zero change records.	booking_change.booking_id → booking.booking_id
Flight - Booking Change (old)	1 : 0..N The old flight reference is optional; one flight may be referenced by many or zero change records.	booking_change.old_flight_id → flight.flight_id

5. Relationships and Cardinality (continued)

Relationship	Cardinality / participation	Implementation (FK)
Flight - Booking Change (new)	1 : 0..N The new flight reference is optional; one flight may be referenced by many or zero change records.	booking_change.new_flight_id → flight.flight_id
Booking - Boarding Pass	1 : 0..1 Each boarding pass belongs to one booking; a booking can have at most one boarding pass.	boarding_pass.booking_id → booking.booking_id (UNIQUE)
Booking - Baggage	1 : 0..N Each baggage item belongs to one booking; a booking may have many or zero baggage items.	baggage.booking_id → booking.booking_id
Baggage - Baggage Check	1 : 0..1 Each baggage check belongs to one baggage item; a baggage item can have at most one check record.	baggage_check.baggage_id → baggage.baggage_id (UNIQUE)
Passenger - Baggage Check	1 : 0..N Each baggage check is associated with one passenger; a passenger may have many or zero baggage checks.	baggage_check.passenger_id → passenger.passenger_id
Booking - Baggage Check	1 : 0..N Each baggage check belongs to one booking; a booking may have many or zero baggage checks.	baggage_check.booking_id → booking.booking_id
Passenger - Security Check	1 : 0..N Each security check belongs to one passenger; a passenger may have many or zero security checks.	security_check.passenger_id → passenger.passenger_id

6. Conclusion

The resulting ER model contains exactly the ten tables implemented in the current PostgreSQL schema. Primary keys provide row identity, foreign keys enforce the defined relationships, UNIQUE constraints create optional one-to-one-style links where required, and NOT NULL constraints define mandatory participation on child rows. The ER diagram and relationship table therefore represent the current database structure without adding entities or attributes outside the implemented schema.

Schema consistency note: The textual descriptions in this report are based on the current executed PostgreSQL definition, including booking_change.old_flight_id and booking_change.new_flight_id and the UNIQUE foreign keys used by boarding_pass and baggage_check.